

Discrete Mathematics

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1. p 16. ex 39 e) f)

39. Construct a truth table for each of these compound propositions.

- a) $p \rightarrow (\neg q \vee r)$
- b) $\neg p \rightarrow (q \rightarrow r)$
- c) $(p \rightarrow q) \vee (\neg p \rightarrow r)$
- d) $(p \rightarrow q) \wedge (\neg p \rightarrow r)$
- e) $(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$
- f) $(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$

e) $(p \leftrightarrow q)$ is true when p and q are equal

$(\neg q \leftrightarrow r)$ is true when $\neg q$ and r are equal

f) $(\neg p \leftrightarrow \neg q)$ is true

when $\neg p$ and $\neg q$ are equal

The same with $(q \leftrightarrow r)$

Finally, if the two expressions are true, the final result will be true, otherwise, false.

Truth table

P	Q	R	$(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$	$(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$
T	T	T	T	T
T	T	F	T	F
T	F	T	T	T
T	F	F	F	F
F	T	T	F	F
F	T	F	T	T
F	F	T	T	F
F	F	F	T	T

2.p 38 ex 5

5. Use a truth table to verify the distributive law
 $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.

This means that both expressions must have the same truth values for all values

Truth table

P	Q	r	$q \vee r$	$P \vee (q \vee r)$	$P \wedge q$	$P \wedge r$	$(P \wedge q) \vee (P \wedge r)$	(Same?)
T	T	T	T	T	T	T	T	Y
T	T	F	T	T	T	F	T	Y
T	F	T	T	T	F	T	T	Y
T	F	F	F	F	F	F	F	Y
F	T	T	T	F	F	F	F	Y
F	T	F	T	F	F	F	F	Y
F	F	T	T	F	F	F	F	Y
F	F	F	F	F	F	F	F	Y

This confirms the distributive law of conjunction over disjunction

3. p 36 ex 33

33. Show that $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ is a tautology.

We need to create the truth table and verify that is always true

Truth table

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$(p \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	T	F	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	T	F	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

It is a tautology

4. p 37 ex 34

34. Show that $(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$ is a tautology.

The same methodology as before

Truth table

p	q	r	$p \vee q$	$\neg p \vee r$	$q \vee r$	$(p \vee q) \wedge (\neg p \vee r)$	$(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	T	F	T
T	F	T	T	T	T	T	T
T	F	F	T	F	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	F	T	T	F	T
F	F	F	F	T	F	F	T

It is a tautology.

5. p 39 ex 55

*55. Find a compound proposition logically equivalent to $p \rightarrow q$ using only the logical operator $\downarrow$.

NAND ($\downarrow$) is defined as $\neg(p \wedge q)$

The standard logical equivalence of implication is $p \rightarrow q \equiv \neg p \vee q$ so, we need to express it only using NAND.

a) Negation: $\neg p \equiv p \downarrow p$

b) OR: $p \vee q \equiv \neg p \downarrow \neg q$

Finally: $p \rightarrow q \equiv \neg p \vee q \equiv (p \downarrow p) \downarrow q$

6. p 3q ex 65

65. Determine whether each of these compound propositions is satisfiable.

- a) $(p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q)$
- b) $(p \rightarrow q) \wedge (p \rightarrow \neg q) \wedge (\neg p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$
- c) $(p \leftrightarrow q) \wedge (\neg p \leftrightarrow q)$

We need to check

if there is at least one assignment that makes the expression true.

a) Truth table

$$\begin{aligned} &(p \vee \neg q) \\ &\quad \wedge \\ &(\neg p \vee q) \\ &\quad \wedge \\ &(\neg p \vee \neg q) \end{aligned}$$

p	q	$p \vee \neg q$	$\neg p \vee q$	$\neg p \vee \neg q$	
T	T	T	T	F	F
T	F	T	F	F	F
F	T	F	T	T	F
F	F	T	T	T	T

Satisfiable

b) Truth table

Unsatisfiable

p	q	$\neg p \vee q$	$\neg p \vee \neg q$	$p \vee q$	$p \vee \neg q$	expression
T	T	T	F	T	T	F
T	F	F	T	T	T	F
F	T	T	T	T	T	F
F	F	T	T	T	T	F

c) Truth table

p	q	$p \rightarrow q$	$\neg p \rightarrow q$	expression
T	T	T	F	F
T	F	F	T	F
F	T	T	T	F
F	F	T	F	F

Unsatisfiable

7. p84 ex 33

33. Use resolution to show that the compound proposition $(p \vee q) \wedge (\neg p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee \neg q)$ is not satisfiable.

1. Convert to normal form

$$C_1 \rightarrow (p \vee q) \quad C_3 \rightarrow (p \vee \neg q)$$

$$C_2 \rightarrow (\neg p \vee q) \quad C_4 \rightarrow (\neg p \vee \neg q)$$

2. Apply resolution (combining 2 clauses that contain complementary literals).

2.1. C_1 and $C_2 \rightarrow (p \vee q) \text{ and } (\neg p \vee q) \rightarrow (q)$

2.2. C_3 and $C_4 \rightarrow (p \vee \neg q)$ and $(\neg p \vee \neg q) \rightarrow (\neg q)$

2.3. Resolve the new clauses $\rightarrow C_4$ and $(\neg q) \rightarrow \emptyset$

We derived an empty clause, this means that the original is unsatisfiable.

8. p 197 ex 8

8. a) Define the domain, codomain, and range of a function.
b) Let $f(n)$ be the function from the set of integers to the set of integers such that $f(n) = n^2 + 1$. What are the domain, codomain, and range of this function?

a) Domain: All possible input values

Codomain: All possible outputs

Range: Actual output values (a subset of codomain)

b) $f(n) = n^2 + 1$

Domain: Since the function takes integers as inputs, the domain is $\mathbb{Z}$

Codomain: The function maps to the set of integers, so the codomain is $\mathbb{Z}$

Range: Since n^2 is always non-negative, the function produces the set greater than or equal to 1. $\{K \in \mathbb{Z} \mid K \geq 1\}$

9. p 147 ex 13

13. Suppose that $a_n = a_{n-1} - 5$ for $n = 1, 2, \dots$. Find a formula for a_n .

$$a_1 = a_0 - 5$$

$$a_2 = a_1 - 5 = a_0 - 5 - 5 = a_0 - 10$$

$$a_3 = a_2 - 5 = a_0 - 5 - 5 - 5 = a_0 - 15$$

We can generalize: $a_n = a_0 - 5n$

Verify by induction:

1. Base case ($n=1$)

$$a_1 = a_0 - 5(1) = a_0 - 5$$

2. Inductive ($n=k+1$)

$$\begin{aligned} a_{k+1} &= a_k - 5 = a_0 - 5k - 5 = \\ &= a_0 - 5(k+1) \end{aligned}$$

10.p 290 ex 41

41. Use the extended Euclidean algorithm to express $\gcd(26, 91)$ as a linear combination of 26 and 91.

We want to find x and y such that:

$$\gcd(26, 91) = 26x + 91y$$

1. $91 \div 26 = 3$; Remainder $= 91 - 3(26) = 13$

$$91 = 3(26) + 13$$

$$26 \div 13 = 2$$
 ; Remainder $= 26 - 2(13) = 0$

Since the remainder is 0, $\gcd(26, 91) = 13$

2. We now express 13 as a linear combination of 26 and 91

$$\text{Since } \gcd(26, 91) = 13 = 91 - 3(26)$$

$$x = -3 \quad ; \quad y = 1$$

11. p 301 ex 6 c) d)

6. Find an inverse of a modulo m for each of these pairs of relatively prime integers using the method followed in Example 2.

a) $a = 2, m = 17$

b) $a = 34, m = 89$

c) $a = 144, m = 233$

d) $a = 200, m = 1001$

c) $\gcd(144, 233)$ using division

- $233 \div 144 = 1$ Remainder $233 - 1(144) = 89$
- $144 \div 89 = 1$ Remainder $144 - 1(89) = 55$
- $89 \div 55 = 1$ Remainder $89 - 1(55) = 34$
- $55 \div 34 = 1$ Remainder $55 - 1(34) = 21$
- $34 \div 21 = 1$ Remainder $34 - 1(21) = 13$
- $21 \div 13 = 1$ Remainder $21 - 1(13) = 8$
- $13 \div 8 = 1$ Remainder $13 - 1(8) = 5$
- $8 \div 5 = 1$ Remainder $8 - 1(5) = 3$
- $5 \div 3 = 1$ Remainder $5 - 1(3) = 2$
- $3 \div 2 = 1$ Remainder $3 - 1(2) = 1$

They are relatively prime, we can proceed with back-substitution.

Starting from: $1 = 3 - 1(2)$

- $(2 = 5 - 1(3))$:

$$1 = 3 - 1(5 - 1(3)) = 2(3) - 1(5)$$

- $(3 = 8 - 1(5))$:

$$1 = 2(8 - 1(5)) - 1(5) = 2(8) - 3(5)$$

- $(5 = 13 - 1(8))$:

$$1 = 2(8) - 3(13 - 1(8)) = 5(8) - 3(13)$$

- $1 = 5(34) - 8(21)$

- $1 = 13(55) - 21(34)$

- $1 = 34(144) - 84(233)$

The inverse of 144 modulo 233 is

$$34 \bmod 233$$

d) Euclidean Algorithm

$$1001 \div 200 = 5 \quad \text{Remainder} = 1001 - 5(200) = 1$$

We stop and express 1 as:

$$1 = 1001 - 5(200)$$

The inverse of 200 modulo 1001 is

$$-5 \equiv 996 \pmod{1001}$$

12. p 325 ex 15

15. Use Fermat's little theorem to evaluate $9^{200} \pmod{19}$.

Fermat's little theorem says that if p is a prime number and a is not divisible by p then: $a^{p-1} \equiv 1 \pmod{p}$

- $p = 19$ (prime)
- $a = 9$ (not divisible)

$$\text{Thus: } 9^{18} \equiv 1 \pmod{19}$$

We can reduce the exponent $200 \bmod 18$

$$200 \div 18 = 11 \text{ remainder } 2$$

$$200 \equiv 2 \bmod 18$$

This means: $a^{200} \equiv a^2 \bmod 19$

• Compute $a^2 \bmod 19$ $a^2 = 81$

$$81 \div 19 = 4 \text{ remainder } 5$$

So: $81 \equiv 5 \bmod 19$

$$a^{200} \equiv 5 \bmod 19$$

13. p 351 ex 15

15. Prove that for every positive integer n ,

$$1 \cdot 2 + 2 \cdot 3 + \dots + n(n+1) = n(n+1)(n+2)/3.$$

1. Case base

For $n=1$, left hand side: $1 \cdot 2 = 2$

Right hand side:

$$\frac{1 + (1+1)(1+2)}{3} = \frac{6}{3} = 2 \quad \rightarrow \quad 2 = 2$$

2. We need to prove also for $n = k+1$

$$1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1) + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$$

3. Inductive step

$$\frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

• Factor out $(k+1)(k+2)$

$$\begin{aligned} \frac{k(k+1)(k+2)}{3} + \frac{3(k+1)(k+2)}{3} &= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3} \\ &= \frac{(k+1)(k+2)(k+3)}{3} \end{aligned}$$

← the formula for $n = k+1$

14. p 418 ex 32

32. How many strings of eight uppercase English letters are there
- if letters can be repeated?
 - if no letter can be repeated?
 - that start with X, if letters can be repeated?
 - that start with X, if no letter can be repeated?
 - that start and end with X, if letters can be repeated?
 - that start with the letters BO (in that order), if letters can be repeated?
 - that start and end with the letters BO (in that order), if letters can be repeated?
 - that start or end with the letters BO (in that order), if letters can be repeated?

- a) Each position in the 8-letter can be any of the 26 letters, since repetition is allowed:

$$26^8$$

- b) For the first letter we have 26 choices, for the second 25 and go on:

$$26 \cdot 25 \cdot 24 \cdot \dots \cdot 19$$

This is a permutation:

$$P(26, 8) = \frac{26!}{(26-8)!} = \frac{26!}{18!}$$

- c) The first letter is fixed as X, the remaining 7 can be any of the 26

$$26^7$$

d) The same as b but we start from 25 and 7 at a time.

$$P(25, 7) = \frac{25!}{18!}$$

e) First and last fixed as X so, the remaining 6:
 26^6

f) The same as e but instead of first and last, first and second.
 26^6

g) We fixed 4 letters, so the remaining 4:
 26^4

h) Includes strings that:
• Start with BO : 26^6

- Finished with 30: 26^6

- Start and finish with 30: 26^4

By the exclusion-inclusion principle:

$$26^6 + 26^6 - 26^4$$

15. p 419 ex 55

55. How many positive integers not exceeding 100 are divisible either by 4 or by 6?

- By 4: They form an arithmetic sequence

First term $\rightarrow a_1 = 4$

Difference $\rightarrow d = 4$

Last term $\rightarrow a_n = 100$

To find n : $4n = 100 \rightarrow n = \frac{100}{4} = 25$ numbers

- By 6: The same:

$a_1 = 6$; $d = 6$; $a_n = 96$

$6n = 96 \rightarrow n = \frac{96}{6} = 16$ numbers

- By both 4 and 6 $\rightarrow 12$: The same:

$$a_1 = 12 ; d = 12 ; a_n = 96$$

$$12n = 96 \rightarrow n = \frac{96}{12} = 8 \text{ numbers}$$

- Apply inclusion-exclusion principle:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$\text{where : } |A| = 25 ; |B| = 16 ; |A \cap B| = 8$$

$$|A \cup B| = 25 + 16 - 8 = 33 \text{ numbers}$$

16. p 428 ex 46

46. There are 51 houses on a street. Each house has an address between 1000 and 1099, inclusive. Show that at least two houses have addresses that are consecutive integers.

- 51 houses
- The total numbers of addresses:
 $1099 - 1000 + 1 = 100$

Pigeonhole: Each represents a pair of

consecutive numbers. (50 pairs)

Principle: We have 51 houses and 50 pairs, by the pigeonhole principle, at least one of these pigeonhole must contain two houses at least.

This means that at least two houses must have consecutive addresses.

17. p 435 ex 25

25. How many ways are there for four men and five women to stand in a line so that

- a) all men stand together?
- b) all women stand together?

a) Since there are 4 men, they can be arranged among themselves in:

$$4! = 24$$

Now we have 6 units: 1 unique unit for men and 5 for women.

These 6 units can be arranged in: $6! = 720$

Since the men can be arranged within their block in $4!$ ways and the 6 groups can be arranged in $6!$ ways, the total number is:

$$6! \cdot 4! = 720 \cdot 24 = 17280$$

b) The same but with women:

$$5! = 120$$

5 total units: 1 woman and 4 men

$$5! = 120$$

Final calculation:

$$5! \cdot 5! = 120 \cdot 120 = 14400$$

18. p 436 ex 30

30. A professor writes 40 discrete mathematics true/false questions. Of the statements in these questions, 17 are true. If the questions can be positioned in any order, how many different answer keys are possible?

Since we are selecting 17 positions out of 40, the number of ways to do this is given by the binomial coefficient:

$$\binom{40}{17} = \frac{40!}{17!(40-17)!} = \frac{40!}{17!23!}$$

This represents the total ways to assign 17 true and 23 false among 40 questions

19. p 436 ex 32

32. Seven women and nine men are on the faculty in the mathematics department at a school.

- How many ways are there to select a committee of five members of the department if at least one woman must be on the committee?
- How many ways are there to select a committee of five members of the department if at least one woman and at least one man must be on the committee?

a) We use the complement principle:

- Total ways with no restriction

$$\binom{16}{5} = \frac{16!}{5!11!} = 4368$$

- Ways with no women

$$\binom{9}{5} = \frac{9!}{5! 4!} = 126$$

- Apply the complement principle

$$\binom{16}{5} - \binom{9}{5} = 4368 - 126 = 4242$$

b) The same but we add with only women:

$$\binom{7}{5} = \frac{7!}{5! 2!} = 21$$

- Apply the complement principle:

$$\binom{16}{5} - \binom{9}{5} - \binom{7}{5} = 4368 - 126 - 21 = 4221$$

20. p 436 ex 46

*46. How many ways are there for a horse race with four horses to finish if ties are possible? [Note: Any number of the four horses may tie.]

• Compute possible tie groupings:

a) No ties: $4! = 24$

b) 3 tie, one separate: $\binom{4}{3} \cdot 2! = 4 \cdot 2 = 8$

c) 3 tie, two separate: $\binom{4}{2} \cdot 3! = 6 \cdot 6 = 36$

d) 2 pairs of tie: $\frac{\binom{4}{2} \cdot 2!}{2} = \frac{6 \cdot 2}{2} = 6$

e) All tie: 1

• Sum all cases: $24 + 8 + 36 + 6 + 1 = 75$

21. p 444 ex 26

26. Prove the identity $\binom{n}{r} \binom{r}{k} = \binom{n}{k} \binom{n-k}{r-k}$, whenever n , r , and k are nonnegative integers with $r \leq n$ and $k \leq r$,

- using a combinatorial argument.
- using an argument based on the formula for the number of r -combinations of a set with n elements.

a) 1. LHS Interpretation

First, choose r elements from a set of n elements. This can be done in $\binom{n}{r}$ ways, then, from those r elements, choose k , this

can be done in $\binom{r}{k}$ ways. This process results in first selecting an r -element subset and then a k -element subset from it

2. RHS Interpretation

First choose k elements directly from the n elements. Now, from the remaining $n-k$ elements choose $r-k$ elements to complete a total of r . This process directly constructs the same selection of k elements and then expands it to r by choosing additional elements

Since both sides count the number of ways to form an r -element subset with an intermediate step of choosing k elements the identity holds.

$$b) \binom{n}{r} = \frac{n!}{r!(n-r)!} ; \binom{r}{k} = \frac{r!}{k!(r-k)!} ;$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} ; \binom{n-k}{r-k} = \frac{(n-k)!}{(r-k)!(n-r)!}$$

• LHS calculation:

$$\binom{n}{r} \binom{r}{k} = \left(\frac{n!}{r!(n-r)!} \right) \left(\frac{r!}{k!(r-k)!} \right) = \frac{n!}{k!(r-k)!(n-r)!}$$

• RHS calculation:

$$\binom{n}{k} \binom{n-k}{r-k} = \left(\frac{n!}{k!(n-k)!} \right) \cdot \left(\frac{(n-k)!}{(r-k)!(n-r)!} \right) =$$

$$= \frac{n!}{k!(r-k)!(n-r)!}$$

Since LHS and RHS are equal, the identity is proven.

22. p 455 ex 14

14. How many solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 = 17,$$

where x_1, x_2, x_3 , and x_4 are nonnegative integers?

The formula for distributing n identical objects (stars) among k distinct groups (variables) is:

$$\binom{n+k-1}{k-1} \quad \begin{array}{l} n=17 \\ k=4 \end{array}$$

$$\binom{17+4-1}{4-1} = \binom{20}{3}$$

Computing $\binom{20}{3}$:

$$\binom{20}{3} = \frac{20!}{3!(20-3)!} = \frac{6840}{6} = 1140$$